NARANJO SCORING WORKSHEET AND CAUSALITY JUSTIFICATION REPORT

WORKORA Synthetic Pharmacovigilance Training Pack | 20 fictional cases

Prepared for: Educational pharmacovigilance assignment

Scope: Review of de-challenge and re-challenge evidence; application of the 10-question Naranjo Adverse Drug Reaction Probability Scale; case-wise scoring and clinical rationale.

Important educational-use statement

All cases and identifiers in the source pack are synthetic. This document is an educational analysis only and is not intended for clinical decision-making, patient care, or regulatory submission. The Naranjo scale supports structured causality assessment; it does not prove causation.

Scoring convention used

The supplied dataset's Expectedness Reference value “Known” was operationally mapped to Question 1 = Yes (+1), while “Unexpected” was mapped to Question 1 = No (0). This is a dataset-based proxy only: the pack does not provide literature citations to independently verify previous conclusive reports. Question 2 was scored Yes (+2) because each case explicitly places the reaction after treatment initiation. For Question 3, only a structured DE challenge value of “Positive” was scored Yes (+1). For Question 4, “Positive” = Yes (+2), “Negative” = No (□1), and “Not done” = Unknown (0). Questions 5–10 were scored Unknown (0) because the pack does not provide sufficient evidence to answer them reliably.

Interpretation bands: Definite □9; Probable 5–8; possible 1–4; Unlikely □0.

1. Naranjo scoring worksheet – all cases

Q1 prior conclusive reports; Q2 event after administration; Q3 improvement on withdrawal; Q4 recurrence on re-administration; Q5 alternative causes; Q6 placebo; Q7 toxic drug level; Q8 dose-response; Q9 similar prior reaction; Q10 objective evidence.

Case ID	Suspect drug / reaction	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total	Category
SYN-PV-011	Metformin / Nausea	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-012	Amlodipine / Peripheral edema	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-013	Ciprofloxacin / Tendon pain	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-014	Sertraline / Dizziness	1	2	0	-1	0	0	0	0	0	0	2	Possible
SYN-PV-015	Omeprazole / Headache	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-016	Losartan / Fatigue	0	2	1	-1	0	0	0	0	0	0	2	Possible
SYN-PV-017	Amoxicillin / Rash	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-018	Atorvastatin / Myalgia	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-019	Metformin / Nausea	1	2	1	-1	0	0	0	0	0	0	3	Possible
SYN-PV-020	Amlodipine / Peripheral edema	1	2	0	0	0	0	0	0	0	0	3	Possible
SYN-PV-021	Ciprofloxacin / Tendon pain	1	2	1	0	0	0	0	0	0	0	4	Possible
SYN-PV-022	Sertraline / Dizziness	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-023	Omeprazole / Headache	1	2	0	-1	0	0	0	0	0	0	2	Possible
SYN-PV-024	Losartan / Fatigue	0	2	1	2	0	0	0	0	0	0	5	Probable
SYN-PV-025	Amoxicillin / Rash	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-026	Atorvastatin / Myalgia	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-027	Metformin / Nausea	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-028	Amlodipine / Peripheral edema	1	2	0	-1	0	0	0	0	0	0	2	Possible
SYN-PV-029	Ciprofloxacin / Tendon pain	1	2	0	2	0	0	0	0	0	0	5	Probable
SYN-PV-030	Sertraline / Dizziness	1	2	1	2	0	0	0	0	0	0	6	Probable

Score summary: Definite: 0; Probable: 13; Possible: 7; Unlikely: 0

No case reached the definite band under the available documentation. The scores are constrained by missing information, especially alternative causes, placebo response, drug concentrations, dose-response, prior patient history, and objective confirmation.

2. De-challenge and re-challenge review

In this dataset, de-challenge and re-challenge are provided as categorical fields. A positive de-challenge supports causality when the reaction improves after withdrawal; a negative de-challenge means improvement was not documented or did not occur. A positive re-challenge supports causality when the same reaction recurs after the drug is restarted. Re-challenge should not be deliberately performed solely to establish causality when it may expose a patient to harm.

Data-quality caution: Some narrative summaries say the patient was "recovering" after treatment was stopped, while the structured DE challenge field is "Negative" or "Unknown." The scoring worksheet follows the structured field to avoid silently changing source data. The mismatch is explicitly noted in the case rationale where applicable.

3. Case-wise causality justification

SYN-PV-011 — Metformin: Nausea | Total 5 — Probable

Patient/context: Age 23; sex Unknown; indication Hypertension; dose 5 mg once daily by Oral; onset 19 day(s) after initiation. Concomitant medicine(s): None reported. Reported seriousness: Medically important.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured DE challenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-012 — Amlodipine: Peripheral edema | Total 5 — Probable

Patient/context: Age 24; sex Female; indication Hypertension; dose 5 mg once daily by Oral; onset 4 day(s) after initiation. Concomitant medicine(s): None reported. Reported seriousness: Medically important.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured DE challenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-013 — Ciprofloxacin: Tendon pain | Total 5 — Probable

Patient/context: Age 75; sex Female; indication Diabetes; dose 500 mg twice daily by Oral; onset 5 day(s) after initiation. Concomitant medicine(s): Vitamin D. Reported seriousness: Medically important.
De-challenge: Negative structured de-challenge: improvement after withdrawal is not supported by the structured field (0).

Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source discrepancy: the narrative describes recovery/recovering after stopping treatment, but the structured DE challenge field is Negative. The structured field was retained for scoring; the discrepancy should be clarified with the source.

SYN-PV-014 — Sertraline: Dizziness | Total 2 — Possible

Patient/context: Age 75; sex Female; indication Diabetes; dose 5 mg once daily by Oral; onset 9 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Hospitalization.
De-challenge: Negative structured de-challenge: improvement after withdrawal is not supported by the structured field (0).
Re-challenge: Negative re-challenge: recurrence is recorded as absent (□1), which weighs against causality.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 -1; Q5–Q10 = 0 each (unknown/not documented). Total
= **2** □ **Possible**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The available Withdrawal/re w administration evidence does not add positive support, or is insufficient. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source discrepancy: the narrative describes recovery/recovering after stopping treatment, but the structured DE challenge field is Negative. The structured field was retained for scoring; the discrepancy should be clarified with the source.

SYN-PV-015 — Omeprazole: Headache | Total 5 — Probable

Patient/context: Age 33; sex Male; indication Hypertension; dose 20 mg once daily by Oral; onset 16 day(s) after initiation. Concomitant medicine(s): None reported. Reported seriousness: Hospitalization.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured DE challenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-016 — Losartan: Fatigue | Total 2 — Possible

Patient/context: Age 66; sex Unknown; indication Diabetes; dose 20 mg once daily by Oral; onset 10 day(s) after initiation.

Concomitant medicine(s): None reported. Reported seriousness: Not serious.
De-challenge: Positive structured de-challenge: improvement after withdrawal is recorded (+1), supporting a temporal association.
Re-challenge: Negative re-challenge: recurrence is recorded as absent (□1), which weighs against causality.
Scoring: Q1 +0; Q2 +2 (event followed drug administration); Q3 +1; Q4 -1; Q5–Q10 = 0 each (unknown/not documented). Total
= **2** □ **Possible**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The supplied expectedness reference is unexpected and contributes no Question 1 point. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

SYN-PV-017 — Amoxicillin: Rash | Total 5 — Probable

Patient/context: Age 57; sex Unknown; indication Hypertension; dose 5 mg once daily by Oral; onset 21 day(s) after initiation. Concomitant medicine(s): Vitamin D. Reported seriousness: Medically important.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured DE challenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-018 — Atorvastatin: Myalgia | Total 5 — Probable

Patient/context: Age 56; sex Female; indication Anxiety; dose 500 mg twice daily by Oral; onset 17 day(s) after initiation. Concomitant medicine(s): None reported. Reported seriousness: Hospitalization.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured DE challenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-019 — Metformin: Nausea | Total 3 — Possible

Patient/context: Age 29; sex Unknown; indication Anxiety; dose 5 mg once daily by Oral; onset 9 day(s) after

initiation. Concomitant medicine(s): None reported. Reported seriousness: Not serious.
De-challenge: Positive structured de-challenge: improvement after withdrawal is recorded (+1), supporting a temporal association.
Re-challenge: Negative re-challenge: recurrence is recorded as absent (□1), which weighs against causality.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +1; Q4 -1; Q5–Q10 = 0 each (unknown/not documented). Total
= **3** □ **Possible**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

SYN-PV-020 — Amlodipine: Peripheral edema | Total 3 — Possible

Patient/context: Age 56; sex Unknown; indication Diabetes; dose 5 mg once daily by Oral; onset 7 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Hospitalization.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Re-challenge was not done or is unknown; no score is awarded (0).
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +0; Q5–Q10 = 0 each (unknown/not documented). Total
= **3** □ **Possible**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The available Withdrawal/re w administration evidence does not add positive support, or is insufficient. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured Dechallenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-021 — Ciprofloxacin: Tendon pain | Total 4 — Possible

Patient/context: Age 33; sex Unknown; indication Infection; dose 5 mg once daily by Oral; onset 9 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Medically important.
De-challenge: Positive structured de-challenge: improvement after withdrawal is recorded (+1), supporting a temporal association.
Re-challenge: Re-challenge was not done or is unknown; no score is awarded (0).
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +1; Q4 +0; Q5–Q10 = 0 each (unknown/not documented). Total
= **4** □ **Possible**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenge and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

SYN-PV-022 — Sertraline: Dizziness | Total 5 — Probable

Patient/context: Age 43; sex Female; indication Infection; dose 5 mg once daily by Oral; onset 21 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Not serious.
De-challenge: Negative structured de-challenge: improvement after withdrawal is not supported by the structured field (0).

Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source discrepancy: the narrative describes recovery/recovering after stopping treatment, but the structured Dechallenge field is Negative. The structured field was retained for scoring; the discrepancy should be clarified with the source.

SYN-PV-023 — Omeprazole: Headache | Total 2 — Possible

Patient/context: Age 26; sex Male; indication Anxiety; dose 500 mg twice daily by Oral; onset 4 day(s) after initiation. Concomitant medicine(s): None reported. Reported seriousness: Hospitalization.
De-challenge: Negative structured de-challenge: improvement after withdrawal is not supported by the structured field (0).
Re-challenge: Negative re-challenge: recurrence is recorded as absent (□1), which weighs against causality.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 -1; Q5–Q10 = 0 each (unknown/not documented). Total
= **2** □ **Possible**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The available Withdrawal/re w administration evidence does not add positive support, or is insufficient. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source discrepancy: the narrative describes recovery/recovering after stopping treatment, but the structured Dechallenge field is Negative. The structured field was retained for scoring; the discrepancy should be clarified with the source.

SYN-PV-024 — Losartan: Fatigue | Total 5 — Probable

Patient/context: Age 79; sex Unknown; indication Hypertension; dose 500 mg twice daily by Oral; onset 3 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Hospitalization.
De-challenge: Positive structured de-challenge: improvement after withdrawal is recorded (+1), supporting a temporal association.
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +0; Q2 +2 (event followed drug administration); Q3 +1; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The supplied expectedness reference is unexpected and contributes no Question 1 point. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

SYN-PV-025 — Amoxicillin: Rash | Total 5 — Probable

Patient/context: Age 61; sex Unknown; indication Infection; dose 500 mg twice daily by Oral; onset 21 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Medically important.
De-challenge: Negative structured de-challenge: improvement after withdrawal is not supported by the structured field (0).

Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source discrepancy: the narrative describes recovery/recovering after stopping treatment, but the structured Dechallenge field is Negative. The structured field was retained for scoring; the discrepancy should be clarified with the source.

SYN-PV-026 — Atorvastatin: Myalgia | Total 5 — Probable

Patient/context: Age 48; sex Unknown; indication Hypertension; dose 20 mg once daily by Oral; onset 12 day(s) after initiation. Concomitant medicine(s): Vitamin D. Reported seriousness: Medically important.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured Dechallenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-027 — Metformin: Nausea | Total 5 — Probable

Patient/context: Age 24; sex Unknown; indication Hypertension; dose 20 mg once daily by Oral; onset 14 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Medically important.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** □ **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured Dechallenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-028 — Amlodipine: Peripheral edema | Total 2 — Possible

Patient/context: Age 77; sex Unknown; indication Hypertension; dose 5 mg once daily by Oral; onset 18 day(s) after initiation. Concomitant medicine(s): Vitamin D. Reported seriousness: Not serious.

De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Negative re-challenge: recurrence is recorded as absent (−1), which weighs against causality.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 -1; Q5–Q10 = 0 each (unknown/not documented). Total
= **2** → **Possible**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The available Withdrawal/re administration evidence does not add positive support, or is insufficient. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured Dechallenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-029 — Ciprofloxacin: Tendon pain | Total 5 — Probable

Patient/context: Age 35; sex Unknown; indication Diabetes; dose 5 mg once daily by Oral; onset 13 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Medically important.
De-challenge: De-challenge is recorded as unknown; no score is awarded (0).
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +0; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **5** → **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

Data-quality note: Source limitation: the narrative mentions recovery/recovering after stopping treatment, but the structured Dechallenge field is Unknown. This was not upgraded to a positive score without explicit confirmation that withdrawal caused improvement.

SYN-PV-030 — Sertraline: Dizziness | Total 6 — Probable

Patient/context: Age 82; sex Male; indication Anxiety; dose 20 mg once daily by Oral; onset 17 day(s) after initiation. Concomitant medicine(s): Paracetamol. Reported seriousness: Medically important.
De-challenge: Positive structured de-challenge: improvement after withdrawal is recorded (+1), supporting a temporal association.
Re-challenge: Positive re-challenge: recurrence after re-administration is recorded (+2), providing supportive evidence.
Scoring: Q1 +1; Q2 +2 (event followed drug administration); Q3 +1; Q4 +2; Q5–Q10 = 0 each (unknown/not documented). Total
= **6** → **Probable**.

Clinical rationale: The reported event occurred after exposure, establishing temporality. The recorded positive de-challenges and/or re-challenge add support to the association. However, the pack does not document a systematic evaluation of alternative causes, placebo exposure, serum drug concentrations, dose-response, prior similar reactions in this patient, or objective diagnostic evidence. The known expectedness reference contributes one point only as a dataset proxy and is not independently verified. Therefore, the classification reflects the limited evidence supplied and should not be interpreted as proof that the suspect medicine caused the event.

4. Overall interpretation and limitations

The Naranjo algorithm is a structured tool for estimating the probability that an adverse event is related to a medicine. It is not a substitute for clinical judgement, a complete patient assessment, or a formal regulatory causality review. A score can be affected substantially by missing or uncertain information.

Key limitations in this training pack

- Question 1: "Known" expectedness was used as a proxy for previous conclusive reports, but no published references or product information excerpts were supplied.
- Question 5: Alternative causes cannot be ruled out because the cases provide limited medical history and no detailed differential diagnosis. Therefore, Unknown (0) was used rather than assuming No (+2) or Yes (−1).
- Question 6: No placebo administration information is provided.
- Question 7: No measured drug concentrations or toxic-range evidence is provided.
- Question 8: No dose reduction/escalation or dose-response information is provided.
- Question 9: No patient-specific history of a similar reaction on prior exposure is provided.
- Question 10: No laboratory results, imaging, examination findings, or other objective confirmation is supplied for the reactions.
- The source includes dose/indication combinations that may warrant verification. They were transcribed as provided and were not clinically corrected or used to infer causality.
- Seriousness and expectedness are separate pharmacovigilance concepts and do not themselves determine the Naranjo causality category.

Conclusion

The classifications in this report are provisional educational outputs based strictly on the supplied synthetic case data and the scoring assumptions stated above. Cases with positive re-challenge and/or positive de-challenge receive higher scores, while absent or unknown evidence is not treated as proof. Before any real-world causality conclusion, the reporter should clarify conflicting timelines, obtain relevant clinical records and objective findings, assess plausible alternative causes, and verify the suspect product's known safety information.

End of report